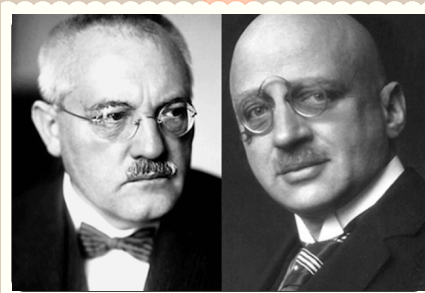




HABER PROCESS

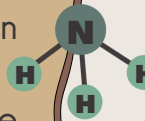


History

The chemical process that killed millions and saved billions

Purpose

The Haber process is used to produce **ammonia** on an industrial scale. It is one of the **most produced** chemical compound in the world and helps feed about **40%** of the world's population.



"In peace-time the scientist belongs to humanity, in war-time to his fatherland"
-Fritz Haber

The Haber process was discovered by **Fritz Haber** and **Carl Bosch** in Germany during WWI.

131 MILLION

metric tons of ammonia are produced a year, 80% used for fertilizer production



CAUTION

High levels of ammonia can cause...

- Skin, mouth, throat, eye irritation.
- Water pollution
- Accidental explosions

Reaction Process



Where the production of ammonia occurs

$\text{H}_2 + \text{N}_2$ Enters here

Reactor

Catalyst

The iron catalyst increases the rate at which a reaction takes place

$\text{H}_2 + \text{N}_2$ Recycled reactants

Coolant Out

Condenser

Coolant In

NH_3

Ammonia is separated from the unused gases

ENVIRONMENTAL IMPACT

- Pollutes groundwater and drinking water
- Bioaccumulate in food chains to toxic levels
- Accounts for 1.2% of the world's total CO₂ emission.



The Haber process is a reversible reaction



It is conducted at 400- 450C



Container operates at the pressure of 200 atm

ECONOMICAL IMPACT

Fertilizers and explosives became substantially cheaper to make since this process didn't require much energy. It also increased agricultural productivity world wide.

APPLICATIONS

This process was used to make:

- Fertilizers,
- Explosives
- Cleaning products
- Nitric acid

CO₂

By Valenzia and Michelle

